The Medium

A Foundational Theory of Reality

Bradley K. DePuy, Independent Researcher

ORCID: 0009-0001-0328-2299

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1 Introduction

1.1 The Core Problem

;; DRAFT language needs refinement

Physics inherited a set of primitives. Object, location, motion, sequential time, the detached external observer. These were abstracted from human-scale perceptual experience the cave dweller's sky, formalized. Every subsequent framework inherited them without examination. Thermodynamics, electromagnetism, relativity, quantum mechanics, string theory, loop quantum gravity. Each built on the same foundation.

The incompatibilities between frameworks GR against QM, the measurement problem, the problem of time are symptoms of foundational mismatch. Not technical failures awaiting mathematical fixes.

1.2 The Language Problem

;; DRAFT language needs refinement

Quantum mechanics borrowed ordinary language and bent each term. Observer no longer means a conscious mind. Measurement no longer means careful observation. Collapse may not be a physical event at all. The borrowed terms carry associations the new meanings cannot support. False familiarity misleads and gatekeeps critique.

The Medium requires its own vocabulary. Three terms replace the inherited ones:

- *Occurrence*: a event the unit of substrate process
- *Registration*: retention of a structural consequence of interaction
- *Boundary*: the interface at which two processes alter each other's state

Chosen to mean exactly what they say. Nothing more.

1.3 A Recognition and a Challenge

Modern science is one of the great achievements of the human mind. Quantum mechanics predicts the behavior of matter at the smallest accessible scales with a precision unmatched in the history of empirical inquiry. General relativity describes the geometry of spacetime and the behavior of gravity across cosmological distances with equal exactness. The Standard Model catalogs the fundamental particles and their interactions with extraordinary completeness. These are not approximations or guesses. They are hard-won structures — the accumulated result of centuries of observation, mathematical invention, and experimental testing.

This document does not dispute any of it.

What it asks instead is the question: *What alternative foundation explains the findings of modern science?* The equations of physics are precise maps. The challenge is to identify the territory they map — not the territory of observable phenomena, which physics has surveyed magnificently, but the foundation that gives rise to a universe in which observation is possible at all.

There is a limitation that physics has not confronted fully. Anyone who has ever done physics is an earth-bound observer, even with the advent of space-borne sensors. That observer arrived late — billions of years after the assumed beginning of the universe — in the midst of an apparently infinite universe. That observer, equipped with sensory tools required for survival, began to ask the questions. Using a dimensional coordinate system (meters, seconds, joules), physics constructed systems defined to understand what is seen from that observation point. Constants, dimensions, and equations carry the fingerprints of that position.

> *Out of an infinite nothing all of creation becomes.*
>
> That sentence is the theory. Everything that follows is the attempt to say it precisely.

2 Abstract

The Medium is a foundational framework prior to the limits of human observation. It cannot be tested or demonstrated directly. It is offered as a starting point upon which others may build.

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The sole primitive is ι: the Medium's act of self-distinction. The Medium distinguishes within itself. ι is that act. Each ι event produces character — the difference the distinction makes. +1 and -1 are the two aspects of one distinction. Character is not a property ι holds. It is what distinguishing is. Without difference there is no distinction. The Medium is the endless occurrence of ι, unconstrained. No boundary, structure, or assumed constraint exists at the substrate level. Space, time, dimensionality, and causality are not conditions imposed upon ι — they emerge from ι-relations. The Planck quantum of action marks the projection boundary between the substrate and observable physics, where one ι event corresponds to $\hbar$.

The central organizing claim of v0.5: processes occur independently. No ordering. No coordination required. Yet existence may impact existence. Sequence is what this looks like from inside an embedded process. The substrate field ϕ records distinction characters across the Medium.

From distinction characters, a three-level algebraic hierarchy is derived by successive combination. At the first level, $\kappa = \sum \iota$ is the coupling character of a local collection — the net distinction character of a group of ι events. At the second level, $= \sum \kappa$ is the total configuration energy of a candidate stable structure. At the third level, projects into spacetime as $\hat{H}$, the Hamiltonian of standard physics — not assumed but derived from substrate structure.

Valid combinations are governed by a logical adjacency constraint: two ι events of identical distinction character produce no new information and therefore cannot occupy adjacent positions in a combination. This is not a physical rule imposed on ι but a logical necessity — identical adjacent distinction characters produce nothing new. The combinatorial space of valid configurations is sharply restricted by this constraint. A minimum threshold of combinatorial complexity must be reached before any stable configuration can form — below this threshold, combinations approach but do not achieve coherence, naturally populating the quantum vacuum as substrate fluctuations.

Stable configurations emerge where reaches a local extremum. These are the substrate analogs of bound states. Mass, charge, and spin are not independent intrinsic properties — they are three projection signatures of a single underlying combinatorial structure. Mass corresponds to N as a first approximation — larger configurations project greater mass. The

specific mass values and the hierarchy between generations are not yet derived. Charge corresponds to the net encounter character of . Spin describes the relational closure of a configuration's encounter pattern within the web. It is not rotation.

The Medium distinguishes three categories of emergent property. Properties emergent from the substrate arise directly from ι-combinations: mass, charge, spin, the quantum vacuum, the Hamiltonian, spacetime, and dimensionality. Properties emergent at the projection layer arise from interactions between stable configurations in spacetime: the fundamental forces, fields, and the full complexity of observable reality. This is the domain of existing physics — The Medium makes no claims there. A third category remains unassigned: color charge, gravity, neutrino oscillation, and the phenomena attributed to dark matter and dark energy have not yet been placed in either layer with confidence. Gravity is the most consequential of these — if spacetime is itself a projection from ι-relations, gravity may bridge both layers in a way the other forces do not.

The four major puzzles of standard cosmology — the entropy problem, the arrow of time, cosmological redshift, and the singular origin of the Big Bang — are addressed in Part III. In each case the puzzle dissolves rather than being solved, because it was generated by a category error that The Medium does not commit: treating projection-layer artifacts as substrate-level facts.

Since ι-arrangements are below the current observational floor, direct testing is not possible with existing instruments. The framework predicts that particle properties treated as intrinsic by standard physics are relational outcomes of substrate combinatorics. New experimental approaches designed to probe transition signatures rather than particle states directly are identified as the path toward indirect verification.

The central open problems are offered as challenges to the theoretical physics community: deriving the specific configurations corresponding to observed particles; establishing which ordered combinations produce which spin values under projection; determining the minimum threshold from first principles; resolving whether color charge and gravity are substrate or projection phenomena; and characterizing dark matter and dark energy candidates within the substrate.

Changes from v0.4: Reorganized around the independence primitive as the central organizing claim. Revised ι: occurrence without attribute; attribute produced at encounter, not carried by ι. Encounter established as the primitive event from which relational character and particle properties derive. Topos theory connection placed at the encounter level. Added Part IV: What The Medium Needs to Prove. Added new vocabulary layer (occurrence, registration, boundary) to replace inherited QM terminology. Added Observer and Observed section dissolving the observer/observed asymmetry at the substrate level. Expanded time section with the real-for-the-observer vs. necessary-for-the-universe distinction. Added The Core Problem and The Language Problem as Introduction subsections.

3 PART I: THE SUBSTRATE

The substrate description is complete in itself. It contains no dimensions, no observers, no physics. It is what it is.

3.1 The Independence Primitive

;; DRAFT central organizing claim of v0.5 language needs refinement

Processes occur independently. No ordering. No coordination required. Yet existence may impact existence.

Sequence is what independence looks like from inside an embedded process. Experienced as before and after. The experience is real. The sequence is not the substrate.

Identity is produced at encounter. Each process occurs on its own terms. The relationships are what the processes are. Remove the web and there is no process. Remove the process and there is no web.

;; DRAFT language needs refinement

Three prior frameworks reach toward this ground and stop short.

String theory places a string at the foundation — a one-dimensional entity carrying vibration modes intrinsically. Each frequency corresponds to a different particle. Spacetime is already there before the string. It is the container. Properties are in the entity, not the encounter.

Loop quantum gravity makes spacetime emergent from a spin network — nodes and edges that are atoms of geometry. That is a genuine step. But the network structure itself is assumed, not derived. LQG does not explain why that relational structure exists. It starts one level too late.

Planck bits — Wheeler's *It from Bit*, Fredkin, Wolfram — place a discrete bit at the foundation carrying 0 or 1 intrinsically. The binary distinction is assumed before any encounter. The question of which value is answered before the question is formed.

ι is none of these. All three prior frameworks place entities first and derive relationships. The Medium places the act of distinction first. Not an entity. Not a web requiring prior connection. The Medium distinguishes within itself. ι is that act. Character is what the act produces.

Planck bits come closest — both are discrete, dimensionless, pre-spatial. The gap is not small. A bit carries its state intrinsically. ι produces character as the act of distinction. That is the difference between assuming distinction and deriving it.

Loop quantum gravity comes closest structurally — both make spacetime emergent from relational structure. But LQG presupposes the relational structure. The Medium derives it from the act of self-distinction.

3.2 The Foundational Claim

The framework rests on a single foundational claim:

> The Medium always is and always was and always will be. It has no beginning and no end. Intrinsic distinction is an eternal property of the Medium — not an event that occurred but a fundamental condition. The structure we observe is not the result of something that happened. It is the expression of the Medium always happening.

This claim does three things at once.

It eliminates the need for an initial starting point. The question of what initiated the universe only arises with the assumption of time. If the Medium always is and always was, there is no before and no starting point. The trigger needed for a beginning dissolves by identifying it as a consequence of a false assumption.

It eliminates time as a primitive. The Medium exists outside of time. What we call time is not a property of the Medium. It is downstream — a consequence of repeated manifestation, not a fundamental feature of reality.

It re-frames differentiation. The emergence of structure is not a process that started. It is a condition that always is. The ground state and the differentiated state coexist within the Medium eternally, because intrinsic distinction — ι — is eternal.

3.3 The Ground State

The Medium at its ground state is a fully autonomous stochastic system. It requires no inputs, no external cause, no temporal framework, and no governing distribution. Its randomness is not randomness within a framework — it is randomness without framework. No privileged outcome. No preferred state.

This is not chaos. Chaos retains structure. The ground state of the Medium precedes structure entirely. It is meta-randomness — randomness that is random about its own nature.

In a medium of infinite extent with no preferred states, every possible configuration has nonzero probability. This is not intuitive but is mathematically unavoidable. The universe is not something that happened against the odds. It is a necessary expression of a medium that has no mechanism for suppressing any possibility.

3.4 Axioms

3.4.1 Axiom I — The Eternal Medium

The Medium always is and always was. It has no beginning, no end, and no external cause.

$$\nexists\, t_0 : M \text{ begins at } t_0 \qquad \nexists\, t_1 : M \text{ ends at } t_1 \tag{1}$$

Status: Axiomatic.

3.4.2 Axiom II — Intrinsic Distinction

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The Medium possesses one property: intrinsic distinction, denoted ι. It is not an operator, not a function, not a process. It is what the Medium is. The Medium distinguishes within itself. ι is that act of self-distinction. Each ι event produces character as the difference the distinction makes. Without difference there is no distinction. Character is not a property ι holds — it is what distinguishing is.

Status: Axiomatic.

3.5 Primitives

3.5.1 Primitive I — The Field

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ι is the Medium's act of self-distinction. Each ι event produces character — the difference the distinction makes. That character is not a property ι holds before the act. It is what the act is. +1 and -1 are the two aspects of one distinction. It has no spatial extent, no duration, no location, no mass, no charge, no spin. It is dimensionless.

Each ι event occupies a relational node — a position defined entirely by its co-manifestation relations. Not by spatial coordinates. Spatial position is what a relational node looks like under projection.

The substrate field ϕ records the distinction character produced by each ι event. M here is the set of relational nodes, not a spatial manifold.

$$\phi : M \to \{+1, -1\} \tag{2}$$

A record of distinction characters. The complete description of the Medium's state.

At the projection layer, one ι event corresponds to a single Planck-scale event — the threshold at which quantum and gravitational effects are simultaneously relevant.

ι events are not driven. No prior state determines which relational node the Medium next distinguishes within itself. They simply occur.

Status: Revised in v0.5.

3.5.2 Primitive II — Distinction and Character

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Each ι event is an act of self-distinction within the Medium. The character it produces — +1 or -1 — is the difference the distinction makes. Not determined in advance. Not driven by prior state. Whether the Medium next distinguishes at a given relational node is not governed. It simply does.

$$\phi(\iota) \in \{+1, -1\} \text{ (character of distinction, no preference, no guarantee)} \tag{3}$$

The substrate imposes no preference on which character is produced. Without differential distinction character, ι events produce only undifferentiated noise. No structure, no stability, no material. Differential character is the logical necessity for material existence.

This is where topos theory bears on the framework. Topos theory allows logic in which a question has no answer until the context that makes the question meaningful exists. Which character a ι event produces is not determined before the event. The question does not exist until the distinction occurs. This is not epistemic. The answer does not exist yet. That is the logical territory topos theory describes, and it is the logical territory ι occupies.

Shared state. No distance. No space. No ordering.

This is prior to locality. Any spatial relationship between x and y is constructed downstream, after structure formation establishes the conditions in which distance becomes meaningful.

Status: Revised in v0.5.

Not a beginning and an end. A breathing.

Because ι is an eternal property of the Medium, this arc does not occur once. It occurs wherever and whenever ι manifests. Encounter patterns form, achieve stability, dissolve — across an infinite relational web that does not notice.

3.6 The Algebraic Hierarchy

The structure of the Medium generates a natural algebraic hierarchy through successive combination. Each level is derived from the level below it; none is assumed.

Level 0 — The Primitive. ι is the sole primitive. The Medium's act of self-distinction. Each ι event produces character — the difference the distinction makes. +1 and -1 are the two aspects of one distinction event. Dimensionless. Pre-spatial. Pre-temporal.

Level 1 — Coupling Character. For a local collection of ι events, the coupling character κ is defined as their sum:

$$\kappa = \sum_i \iota_i \tag{4}$$

κ is the net distinction character of a local cluster. It is the first derived quantity in the hierarchy.

Level 2 — Configuration Energy. For a candidate stable structure composed of multiple κ clusters, the total configuration energy is:

$$= \sum_j \kappa_j \tag{5}$$

 governs whether a configuration achieves stability. Stable configurations are those for which reaches a local extremum — a minimum or maximum in the combinatorial landscape of valid arrangements. These are the substrate analogs of bound states. All observed stable particles are candidates for such extrema.

Level 3 — The Hamiltonian. Under projection into the observer's spacetime coordinate system, becomes $\hat{H}$, the Hamiltonian of standard quantum mechanics:

$$\hat{H} = \mathrm{proj}() \tag{6}$$

The Hamiltonian is not an assumption of the theory. It is a derived consequence of the substrate structure, projected through the observational boundary.

The Adjacency Constraint. Two relational nodes are adjacent if their co-manifestation is direct — no third ι event mediates the relation. Valid combinations must satisfy: two ι events of identical distinction character cannot be directly co-manifested. This is not a physical rule but a logical necessity — identical distinction characters in direct co-manifestation produce no new distinction and therefore no new structure. The constraint sharply restricts the combinatorial space and is the source of the minimum complexity threshold below which stable configurations cannot form.

3.7 The Substrate — Complete

The substrate description is now complete. It contains:

- One substrate: the Medium
- One organizing claim: the independence primitive — processes occur independently; existence may impact existence
- One property: ι — the Medium's act of self-distinction, producing character as the difference the distinction makes
- One field: $\phi \in \{+1, -1\}$ — a record of distinction characters
- One relationship: co-manifestation $x \sim y$
- One algebraic hierarchy: $\iota \to \kappa \to\to \hat{H}$
- One arc: maximum undifferentiation → structure → maximum undifferentiation

  Nothing else belongs here. Everything else is downstream.

4 PART II: THE PROJECTION LAYER

Everything in Part II is downstream of the substrate. Dimensions, constants, observers, and all of physics live here — valid, precise, and complete as descriptions of projected structure. Not descriptions of the substrate itself.

4.1 The Observer's Position

Before describing the projections, it is necessary to name what the observer is and what it is not.

An observer is a self-sustaining structure in the binary field — a stable pattern that persists across transitions, embedded in the differentiated region it constitutes. Every observer who has ever done physics is such a structure. Every human observer is an earth-bound variant: carbon-based, temperature-sensitive, confined to a narrow spatial and temporal window, equipped with sensory apparatus tuned to a small fraction of the available signal, and dependent on cognitive architecture shaped by survival pressures that predate science by orders of magnitude.

These are not complaints. They are structural facts that bear on what physics can and cannot see from its current position.

The observer arrives late. The substrate has been operating — in whatever sense "operating" applies to an eternal, timeless ground — without observers for the overwhelming majority of the arc. The observer arrives embedded in a mature, highly differentiated region of the binary field. It has no access to the substrate prior to structure formation. It has no access to regions beyond the propagation horizon of its own local structure. Its coordinate system — meters, seconds, joules, kelvins — was built to make *its own scale* tractable. That coordinate system is extraordinarily powerful within its domain. It is not a description of the substrate. It is a description of the projection.

The constants of physics — h, c, G, k — are the seams where the observer's dimensional framework meets the substrate's dimensionless operations. They are not brute inputs. They are artifacts of the position from which the measurement is taken.

A different observer, embedded in a different region of the binary field at a different stage of the arc, with a different sensory range and a different coordinate system, would derive different constants and call them fundamental. They would be equally right and equally downstream.

4.2 Physics as Projection

The Medium is pre-dimensional. The binary field ϕ carries no units. The transition index t is a dimensionless label. c does not exist at the substrate level. The Independence Primitive establishes that processes occur independently with no shared temporal framework. A rate requires counting transitions against time. No substrate-level time exists. Therefore no substrate-level rate exists, and c as a transition rate limit is a category error at the substrate level. c is projection-layer emergent — the ratio of projected distance to projected time, fixed by the structure of projection itself.

Dimensions do not exist in the substrate. They are constructed by observers.

To describe the relationships between structures like itself, the observer constructs a coordinate system — a framework of units and dimensions that makes the relationships tractable. That coordinate system is not a discovery about the substrate. It is a construction of the observer.

> Every dimensional quantity in physics — every constant, every observable, every equation — is a projection of a substrate relationship onto the observer's dimensional coordinate system. Physics is the complete and consistent description of those projections. It is a perfect map of the shadow. It is not a description of the substrate that casts it.

4.3 The Projection Boundary

The Planck scale marks the projection boundary: the threshold at which one ι event becomes visible to the observer's dimensional framework. At this boundary, the substrate's dimensionless occurrence acquires the dimensional form the observer calls action — energy multiplied by time.

The constancy of h follows directly. Every ι event is identical — the same occurrence of the same property of the same substrate, without variation. The observer always measures the same thing and always gets the same number. The constancy of h is not a fact about physics. It is a consequence of the irreducible uniformity of ι.

An open question is carried explicitly: why does the projection carry units of action specifically — energy multiplied by time — and not some other dimensional combination? The answer requires a derivation of the observer's dimensional categories from the structure of sustained binary field configurations. That derivation is a primary target for future development.

4.4 Emergent Properties

The Medium distinguishes three categories of emergent property, organized by the level at which they arise.

Substrate emergent. These properties arise directly from ι-combinations, prior to any observer: mass (corresponding to N, the count of ι events in a configuration), charge (corresponding to the net distinction character of), spin (arising from relational closure of the configuration's encounter pattern within the web), the quantum vacuum (substrate fluctuations below the stability threshold), the Hamiltonian (as proj()), spacetime (as the projection framework constructed by embedded observers), and dimensionality (as the coordinate structure observers impose on substrate relationships).

Projection layer emergent. These properties arise from interactions between stable configurations within the spacetime projection: the fundamental forces, quantum fields, and the full complexity of observable physical reality. This is the domain of existing physics. The Medium makes no claims in this domain beyond identifying it as downstream.

Unassigned. The following have not yet been placed with confidence in either layer: color charge, gravity, neutrino oscillation, dark matter, and dark energy. Gravity is the most consequential — if spacetime is itself a projection from ι-relations, gravity may bridge both layers in a way the other forces do not. The current limit of assignment reflects the limit of present understanding, not the limit of The Medium.

4.5 Mass, Charge, and Spin as Projection Signatures

Standard physics treats mass, charge, and spin as independent intrinsic properties of particles. The Medium identifies them as three projection signatures of a single underlying combinatorial structure.

Mass corresponds to N as a first approximation: the total count of ι events in a stable configuration. A heavier particle is a configuration with a larger N. The specific values — why the muon is 207 times the electron's mass, why three generations exist at all — are not yet derived from substrate structure.

Charge corresponds to the net distinction character of . A configuration in which one distinction character dominates projects to positive charge; the opposing character projects to negative charge; exact balance projects to charge neutrality. Charge is not a property added to a particle. It is the distinction character of its substrate configuration.

Spin describes how a stable configuration's encounter pattern closes within the relational web. It is not rotation. The particle does not spin. The word is retained from the observation — discrete deflection in Stern-Gerlach — but the property is relational, not mechanical.

A configuration whose encounter pattern returns to its original relational state after one traversal of the web projects as integer spin. A configuration requiring two traversals projects as half-integer spin. The distinction is not internal to the configuration alone. It is in how the configuration's encounter pattern sits within the surrounding web.

The specific derivation of which encounter patterns produce which spin values (0, 1/2, 1, 3/2, 2) remains an open problem. The claim that spin is a property of relational closure is current.

4.6 Observer and Observed

;; DRAFT language needs refinement

Standard quantum mechanics imports a Newtonian posture. A detached external entity measuring a passive system. That posture breaks down at the quantum scale. It was never properly grounded anywhere.

In The Medium there is no outside. Every interaction is between -level processes within the Medium.

The observed is any process whose boundary state is altered by interaction. The observer is any process that retains a structural consequence of that interaction registration. The asymmetry dissolves. Observation is mutual and structural. Not performed from a privileged external position.

Superposition is not a rapid toggling between states interference patterns from the double slit rule that out definitively. It is the formalism's acknowledgment that before interaction, definite states do not exist. The measurement problem has gone unsolved for a century because the observer/observed distinction was never grounded. The Medium dissolves the distinction at the substrate level. What physics calls measurement is registration. What physics calls observer is any process capable of retaining a structural consequence of boundary contact.

4.7 Co-Manifestation and Entanglement

At the substrate level, two ι events producing the same ϕ character share a state with no spatial relationship between them. There is no distance at the substrate level. There is no *between*.

When an observer embedded in differentiated structure encounters two events that are correlated regardless of their spatial separation, it calls this entanglement. Entanglement is the observer's projection of co-manifestation $x \sim y$ into a framework that expects spatial locality to govern correlation. The apparent mystery of entanglement — correlation without causal connection across space — dissolves at the substrate level, where space does not yet exist and co-manifestation requires no connection at all.

4.8 The Stochastic Character — Observer's Description

At the substrate level, ι events are not driven. No prior state determines which relational node the Medium next distinguishes within itself. Each act of self-distinction produces a character — +1 or -1 — with no preference and no guarantee. An observer describing this stochastic character mathematically recognizes it as a Bernoulli process — a binary outcome with associated probability $p(\iota)$ for each ι event. This is the observer's mathematical framework imposed on the substrate's stochastic character. The substrate does not know about Bernoulli. Distinction simply occurs or does not.

The full statistical mechanics of the ground state — entropy, temperature, distributions — emerges on the observer side as the projection of the substrate's undifferentiated stochastic character into the observer's thermodynamic framework.

4.9 The General Principle

The presence of dimensional constants in the equations of physics — h, c, G, k — is the signature of projection. Each constant marks a place where the observer's dimensional framework has been imposed on a substrate relationship that has none. In the substrate's own terms, those constants do not appear.

The program implied by the symbol table: for each row relating a substrate quantity to an observer quantity, derive the dimensional form of the observer quantity from the substrate quantity and a formal account of the projection boundary. If that program succeeds, every fundamental constant of physics will have been given a substrate derivation — and the mystery of their values resolved, not by finding deeper constants, but by understanding that they were never fundamental to begin with.

4.10 On Existing Physics

The existing mathematical frameworks of physics are valid descriptions of differentiated structure behavior within their respective domains. They are not wrong. They are downstream. They are precise and consistent descriptions of the observer's projection of the substrate.

The Medium framework does not replace them. It identifies the substrate from which they project — and defines the program by which their dimensional constants can, in principle, be derived rather than assumed.

5 PART III: COSMOLOGICAL IMPLICATIONS

The following develops four interconnected consequences that follow directly from the foundational structure of The Medium. Each addresses a major puzzle in standard cosmology. In each case the puzzle dissolves rather than being solved — it was generated by a category error that The Medium does not commit: treating projection-layer artifacts as substrate-level facts.

5.1 Entropy as Emergent Accounting

Standard cosmology inherits a puzzle from Boltzmann: if the second law of thermodynamics is statistical rather than fundamental, why was entropy low at the beginning? The standard response bundles this into initial conditions and calls it the past hypothesis. The Big Bang is asserted to have been an extraordinarily low-entropy state. This assertion is not derived. It is required by the model and inserted by hand.

The Medium does not accept this framing, and not merely because it rejects the Big Bang as an explanation. The more fundamental problem is that the framing assigns entropy as a global intrinsic property of the universe — a quantity that belongs to the cosmos as a whole and increases over cosmic time. This is a category error.

Entropy, as Boltzmann formalized it and Shannon made fully explicit, is a measure of an observer's ignorance about the microstate of a system given knowledge of its macrostate. It is not a property of the system alone. It is a property of the relationship between an observer with limited resolution and a system with more degrees of freedom than the observer can track. Entropy is always relative to a coarse-graining — a choice, made by an observer, of which distinctions count and which do not.

The universe as a whole has no external observer. Assigning it a global entropy value requires an observer standing outside it with a complete accounting of all microstates, which is incoherent. What cosmologists call the entropy of the universe is always the entropy of the observable universe as seen from a particular embedded vantage point with particular resolution limits. It is local. It is relational. It is projection-layer all the way down.

In The Medium, the substrate — the ι-combinatorial structure — has no thermodynamic direction. There is no privileged low-entropy initial condition because there is no initial condition at the substrate level in any cosmologically meaningful sense. The apparent entropy gradient that standard cosmology treats as a cosmic fact is a projection-layer phenomenon: an artifact of embedded observers constructing temporal sequences and tracking accessible macrostates from within a particular slice of configuration space.

Entropy is therefore classified in The Medium as projection-layer emergent. It has no substrate analog.

5.2 Time as Projection and Its Consequences for Frequency

The Medium establishes that time is not a substrate-level property. Temporal directionality — the sense that events occur in sequence, that causes precede effects, that entropy increases in one direction — emerges at the projection layer as observers at a particular scale construct ordered accounts of substrate transitions.

Time is real for the embedded observer. Thermodynamically, experientially, irreversibly real. Dismissing it as illusion is philosophical evasion. But real-for-the-observer and necessary-for-the-universe are different claims entirely. The universe's processes do not require time to coordinate. Gravity does not consult a clock. The gravitational relationship between earth and moon is not a sequence of instructions passed back and forth — it is a continuous configuration of the substrate, of which the 1.3-second propagation delay at the projection layer is the observer's local account. Physics wrote the observer's experience of sequence into the equations as a global feature of reality. The equations keep resisting it. General relativity is time-symmetric. The Wheeler-DeWitt equation has no time variable. Time is emergent downstream of process. Not upstream of it.

This has a direct consequence for the concept of frequency. Frequency is defined as cycles per unit time. If time is projection-layer emergent, then frequency is also projection-layer emergent. It is not a substrate-level property of a propagating pattern. It is a measurement that an observer constructs by counting transition events against a locally projected time reference.

Two observers projecting different temporal frames — because they sit at different configuration depths within the substrate, or because their local ι-configuration densities differ — will assign different frequency values to the same arriving pattern. This is not a Doppler effect. It is not recession. It is a structural consequence of time being a local projection rather than a universal background.

5.3 Redshift as Multi-Layer Projection Artifact

Standard cosmology interprets cosmological redshift as evidence of recession velocity: distant galaxies are moving away, space between them and us is expanding, and their light is stretched to longer wavelengths in proportion to the expansion. This interpretation underlies the Big Bang model, Hubble's law, and the entire framework of an expanding universe with a singular low-entropy origin.

The Medium offers a more parsimonious account that requires no expansion, no recession, and no singular origin — while accepting the observational data entirely.

Within The Medium, what we call a photon is a stable propagating ι-configuration pattern —

a structure that persists and moves through successive co-manifestation relationships across configuration space. What we call speed is a projection-layer quantity: a ratio of projected distance to projected time constructed by the observer. It is not a geometric velocity through space. It is not a substrate-level rate. The substrate has no rates — no shared temporal framework exists against which transitions could be counted.

Redshift arises from three stacked projection effects.

First: the emitting configuration. A source at some region of the substrate produces a transition pattern at a rate determined by local ι-configuration density. That rate is what the source's local projection layer calls frequency.

Second: traversal through configuration depth. The pattern propagates across substrate regions that are not uniform. Configuration density varies across the substrate — not directionally, not as expansion, but structurally, as a feature of the combinatorial landscape. As the pattern traverses this varying density, the relationship between transition events and projected distance is not constant.

Third: the receiving configuration. The observer receives the arriving pattern and measures its frequency against a locally projected time reference. That time reference is constructed from the observer's local ι-configuration density, which differs from the emitter's. The frequency the observer assigns is a ratio constructed entirely within the projection layer, against a local clock that has no substrate-level universality.

The redshift that results from this three-layer process is not evidence that the source is receding. It is evidence that the emitting and receiving configurations sit at different depths within a non-uniform substrate, and that the projection relationship between them is asymmetric. What standard cosmology interprets as Hubble's law — a proportional relationship between distance and recession velocity — is, in The Medium, a projection artifact of sampling along a particular observational cone from a single embedded vantage point through a substrate with a particular large-scale configuration structure.

The mature and fully structured galaxies observed at high redshift — including findings from JWST that strain the standard timeline — are consistent with this account. They do not violate a law of nature. They strain a model built on a locally-derived entropy narrative and a misread projection artifact. In The Medium, structural complexity at any configuration depth is not constrained by elapsed cosmic time, because cosmic time is itself a projection.

5.4 The Big Bang as Projection Error

The Big Bang model rests on three foundational assumptions, each of which The Medium addresses directly.

First: that cosmological redshift indicates recession velocity, implying an expanding universe with a singular dense origin. Second: that entropy has a global cosmic direction, requiring a special low-entropy initial state. Third: that time provides a universal background against which an origin event — a moment of creation — is a coherent concept.

The Medium has shown that all three assumptions are projection-layer artifacts. Redshift is configuration-depth asymmetry misread as velocity. Entropy is a local observer relationship with no substrate reality. Time is a local projection with no substrate universality.

Remove those three assumptions and the Big Bang has no remaining evidential foundation.

The observational data — redshift, the cosmic microwave background, large-scale structure — remain. The interpretation built on top of them does not survive.

This is a stronger position than disproof. Disproof engages a model on its own terms, accepting its conceptual foundations and showing internal inconsistency. What The Medium establishes is more fundamental: the conceptual foundations were never valid. The Big Bang is not a theory that turned out to be wrong. It is a theory that was never well-formed — a projection error that becomes visible the moment one insists on asking what is actually happening at the substrate level.

The universe, in The Medium, has no beginning in any substrate-meaningful sense. It has no privileged center, no singular origin, no global clock, and no cosmic entropy gradient. What it has is a substrate of extraordinary combinatorial richness from which structure, observers, and the local projections we call time, space, and thermodynamics all emerge — consistently, without residue, and without requiring any special initial conditions.

6 PART IV: WHAT THE MEDIUM NEEDS TO PROVE

;; DRAFT language needs refinement

Not a physical theory with testable predictions. Not a mathematical framework with derivational power.

Three things.

First: that the primitives are genuinely primitive. That occurrence, registration, boundary do not smuggle in hidden assumptions. That they do not secretly depend on Newtonian concepts.

Second: that existing frameworks require something like these primitives to be coherent. That the measurement problem, the time problem, the observer confusion all point to a gap sitting exactly where The Medium works. Three frameworks in particular are candidates for derivation from the substrate.

Planck bits are what encounter characters look like to an observer who cannot see below the registration level. Digital physics assumes the bit as primitive. The Medium shows where it comes from. The path from encounter to registered character to bit is the derivation. It is not complete. The direction is clear.

Loop quantum gravity makes spacetime emergent from spin networks — nodes and edges that are atoms of geometry. The Medium's encounter web is already that relational structure. Stable encounter configurations at extrema are the nodes. The encounter relationships between them are the edges. LQG's spin foam — the four-dimensional history of spin networks — is the observer's projection of encounter patterns across what it calls time. The derivation would show that spin networks are what the encounter web looks like when an observer imposes a spatial coordinate system on it.

String theory places vibrating strings at the foundation. Different vibration modes correspond to different particles. In The Medium, different encounter configurations at extrema correspond to different particles. The correspondence is directionally plausible. String theory presupposes spacetime and requires ten or eleven dimensions. The derivation would need to show those dimensions emerging from encounter structure. That is the longest path of the three.

None of these derivations is complete. All three are named targets. Showing that existing frameworks are derivable from the substrate is the substance of this second criterion.

Third: that the framework is internally consistent and generative. That thinking from these primitives produces coherent consequences. That it opens questions existing frameworks could not formulate.

The model is the 1905 paper on special relativity. Not complex. Precise. Spare. The argument careful. The boundaries clearly stated.

That is achievable. And it is enough.

7 PART V: DERIVATIONS

;; DRAFT language needs refinement

7.1 Derivation I: Planck Bits from the Substrate

Digital physics — Wheeler's *It from Bit*, Fredkin, Wolfram — claims the bit as primitive. A binary value, 0 or 1, is the most fundamental thing. Everything physical arises from it. No explanation is offered for where the bit comes from. It simply is.

The Medium derives it.

Step 1. ι occurs. The Medium distinguishes within itself. That is the event. It is not driven. No prior state determines it. It simply occurs.

Step 2. The distinction has two aspects: +1 or -1. Not chosen from outside. Not determined in advance. The character of the distinction itself. Without difference there is no distinction. The character is what distinguishing is.

Step 3. The character is registered in ϕ for that ι event. The field records it. One distinction. One binary value.

Step 4. At the projection boundary, one ι event corresponds to one Planck-scale event. This is the threshold at which the substrate's dimensionless distinction acquires the dimensional form the observer calls action. One ι event. One Planck unit. One registration.

Step 5. What the observer records at the Planck scale is one binary value: +1 or -1.

That binary value is the bit.

A Planck bit is not a primitive. It is the registered distinction character of one ι event, as seen from the observer's position. The bit presupposes the act of distinction. Digital physics correctly identifies the information structure of reality. It misidentifies that structure as the ground floor.

Consequences that follow without additional derivation:

The stochastic character of bits — no prior state determines the next — follows directly from ι events not being driven. No prior state determines where or when the Medium next distinguishes within itself.

The minimum size of a Planck bit — information cannot be compressed below one Planck unit — follows from ι being the unit of distinction. Below one distinction event, there is nothing to register. The Planck scale is not arbitrary. It is the projection boundary of the substrate's irreducible unit.

The adjacency constraint explains why directly co-manifested identical bits produce no new information. Two ι events of identical distinction character in direct co-manifestation produce no new distinction. This is not a rule imposed on the bits. It is a logical consequence of what distinction is.

What is not yet derived:

Why one ι event projects to exactly $\hbar$ of action — energy multiplied by time — and not some other dimensional combination. This remains in unresolved questions. The derivation above stands without it.

7.2 Derivation II: Standard Particles as Planck Bit Configurations

;; DRAFT language needs refinement

The Planck bit derivation has an immediate consequence. If a Planck bit is a registered ι distinction character, then a standard particle is a stable configuration of Planck bits — a pattern of distinction characters that has achieved stability at a extremum.

The properties of that particle are what the pattern projects to the observer.

Mass corresponds to N as a first approximation: the count of ι events in the configuration. A heavier particle is a larger pattern. The lightest particles are the threshold cases — the minimum number of Planck bits that can achieve stability under the adjacency constraint. The specific mass values and the three-generation hierarchy are not yet derived.

Charge corresponds to net distinction character. A pattern in which one character dominates projects as charged. Exact balance projects as neutral.

Spin corresponds to relational closure. The pattern either closes in one traversal of the web or two. One traversal: boson. Two traversals: fermion.

The adjacency constraint shapes the particle table.

Identical adjacent distinction characters produce nothing. Only patterns that vary can build. This sharply limits which configurations are possible. The number of stable particles is not arbitrary. It is constrained by which patterns satisfy the adjacency rule. The particle table may be exactly the set of patterns the constraint permits.

The threshold cases:

The electron is the minimum stable configuration with net character imbalance and half-integer closure. Possibly the simplest stable fermion the adjacency constraint allows.

The photon does not sit at a extremum. It is a propagating alternation of distinction characters — the adjacency constraint satisfied but no bound state achieved. The observer assigns it speed c — a projection-layer ratio, not a substrate property. Why projection fixes that ratio at c and not some other value is an open derivation.

The neutrino approaches the stability threshold from below. Near-zero mass. Balanced character. The minimum fermion-closing configuration.

Quarks cannot satisfy the adjacency constraint alone. Stability requires combination. Confinement is not a force imposed from outside. It is a consequence of which Planck bit patterns can and cannot achieve independent stability.

The open derivation:

Which specific Planck bit patterns — satisfying the adjacency constraint, achieving a extremum — correspond to each observed particle? That is the derivation. It begins with the lightest threshold cases and works upward through the particle table. The framework predicts that the table is finite and complete. The adjacency constraint permits exactly the particles that exist. No derivation yet confirms this. It is the primary structural task of the next stage.

8 Unresolved Questions

These are the frontier of the framework, stated precisely and carried explicitly.

The Vocabulary Boundary. The terms occurrence, registration, and boundary are proposed as genuinely primitive replacements for the inherited vocabulary of observer, measurement, and collapse. Whether they succeed — whether they are themselves free of hidden Newtonian assumptions — is an open question requiring formal examination.

The Topos Theory Connection. operates below Boolean logic. Occurrence as boundary negotiation is not a yes/no event — it is a process whose definiteness is contextual and relational. Topos theory, developed by Grothendieck, allows contextual relational logic where truth values are not simply binary. The Medium arrives at this logical territory from the conceptual direction; topos theory approaches from the mathematical direction. Whether they converge formally is an open derivation.

The Nature of ι. What is intrinsic distinction? The provisional characterization — the property permitting local, spontaneous reduction from maximum undifferentiation in a fully random substrate — is the central open question of the theory.

The Formal Account of the Projection Boundary. How does an observer embedded in the binary field construct a dimensional coordinate system? What determines which dimensional categories emerge? This is the primary task for the next stage of formal development.

Why Does One Manifestation Project to Action? The observer's measurement of one irreducible substrate occurrence produces a quantity with units of action — energy multiplied by time. Why this dimensional combination? Until that derivation exists, the correspondence between one unit of substrate process and h is a named target, not a result.

The Derivation of Each Constant and the Projection Boundary. The framework claims the Planck scale is the projection boundary — the threshold at which one ι event corresponds to one Planck unit. But the Planck scale is defined by three constants: $\hbar$, c, and G. The Planck length is $\sqrt{\hbar G/c^3}$. The Planck time is $\sqrt{\hbar G/c^5}$. Until $\hbar$, c, and G are derived from projection structure, the projection boundary is not derived — it is a conjecture about scale, named and deferred. c is not a substrate property. The Independence Primitive establishes no substrate-level time and therefore no substrate-level rate. c is the ratio of projected distance to projected time, fixed by the structure of projection itself. Why projection fixes that ratio at the value it does is the open derivation. $\hbar$ from the correspondence between one ι event and one unit of action. G from structure formation geometry — itself unresolved because gravity has not yet been placed confidently in either layer. These three are the load-bearing unresolved derivations of the framework. The substrate has no preferred frame — frames arise only at projection. Lorentz invariance is therefore a property of how projection works, not a constraint violated by the substrate. The substrate is pre-frame. That is the resolution of the Lorentz invariance question. The derivation that confirms it — showing projection recovers exact Lorentz invariance — does not yet exist.

Particle Configurations. Standard particles are stable Planck bit configurations at extrema. The adjacency constraint limits which configurations are possible. The particle table may be exactly the set of patterns the constraint permits — finite, complete, and derivable. The open derivation: identify the minimum stable Planck bit patterns satisfying the adjacency constraint and match them to the observed particle table, beginning with the lightest threshold cases. This is the primary structural task of the next stage.

Spin as Relational Closure. Which encounter patterns produce which spin values (0, 1/2, 1, 3/2, 2)? The claim that spin is a property of relational closure — integer spin closing in one traversal of the web, half-integer in two — is current. The derivation of which specific encounter patterns correspond to which values is open. More consequentially: the framework names the topology of spin but does not yet connect it to exchange statistics. Fermions obey Pauli exclusion — no two in the same quantum state. Bosons do not. That difference determines atomic shell structure, the stability of matter, and the behavior of light. The spin-statistics theorem in quantum field theory derives this connection from Lorentz invariance, causality, and positivity of energy. The Medium's relational closure description is consistent with the topology the theorem requires but does not derive the statistics. Doing so requires showing that projection from the substrate recovers Lorentz invariance, from which the spin-statistics theorem follows as a downstream result. That derivation does not yet exist.

Derivation of Prior Frameworks. Three existing frameworks are candidates for derivation from the substrate. Planck bits: the registered encounter character is a bit — the derivation shows where the bit comes from. Loop quantum gravity: the encounter web maps to spin networks; stable configurations map to nodes; encounter relationships map to edges; the derivation shows LQG as the observer's projection of the substrate's relational structure. String theory: different encounter configurations at extrema correspond to different particles as string vibration modes correspond to different particles — the derivation requires showing how the encounter structure generates the dimensional space string theory presupposes. Each is an open derivation. Each is a named target.

Fractional Charge. The framework produces integer net distinction characters from ś1 combinations. Quarks carry charge ś1/3 and ś2/3 — never observed in isolation, always confined within hadrons. The hadronic totals are integers; the quark-level values are not. The ś1 substrate system cannot produce 1/3 as a global value. A possible path: topos theory, already in the framework at the level, may apply here as well. In a non-Boolean topos, local sections can exist that do not extend to global sections. A quark's charge may be a local section over the context "confined in a hadron" — meaningful only within that context, not a global element of the substrate's ś1 system. The same contextual logic that governs which character produces before distinction occurs may govern which charge value a quark carries before isolation is possible. This is a direction, not a derivation. It is carried here as an open connection.

The Mass Hierarchy. Mass corresponds to N as a first approximation. That correspondence does not explain why the electron is 0.511 MeV, the muon 105 MeV, and the tau 1,777 MeV — identical structure, identical charge, identical spin, three wildly different masses. It does not explain why there are exactly three generations. The Standard Model inserts these values by hand. The Medium claims mass is derivable from substrate structure — that claim requires showing which combinatorial configurations produce which specific N counts, and why those counts correspond to the observed values. This derivation does not yet exist.

The Cosmic Microwave Background. The Medium offers an alternative account of cosmological redshift and rejects the Big Bang as a projection error. That position requires an alternative account of the CMB — and none is yet given. The CMB presents three independent constraints any alternative cosmology must satisfy: a near-perfect blackbody spectrum at 2.725 K, requiring a hot opaque early phase; uniformity across causally disconnected regions, requiring an account of how that uniformity arose; and precisely measured anisotropies that match the seeding of large-scale structure. The Medium's redshift account addresses none of these. This is the most consequential open problem in the cosmological section. It is carried explicitly here and deferred to future work.

Color Charge, Gravity, and the Unassigned Category. Whether color charge is substrate or projection-layer emergent; whether gravity bridges both layers through its connection to spacetime; how dark matter and dark energy candidates arise from substrate structure. These are the frontier unresolved questions.

The Observer Boundary. The observer is a sustained binary configuration within ϕ, attempting to describe the substrate that produced it from within. Every formal system contains truths it cannot prove from within itself. The Medium faces an analogous boundary and acknowledges it.

9 Summary

10 A Personal Note

I am seventy-nine years old. I am not an academic. I have no credentials in physics or mathematics. I began this work with a question that would not leave me alone and a conviction that the question deserved a serious attempt at an answer.

I began this knowing I may not live to see it reach its conclusions. That does not trouble me. What matters is that the ideas are documented, attributed, and made available to anyone who finds them worthy of development. If this framework contains something true it belongs to whoever can take it further. My name attached to it is enough — for the record, and for my family.

Gregor Mendel published his work on inheritance and it sat unread for decades before being recognized as foundational. The ideas belong to whoever finds them. The record belongs to their origin.

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Table 1: Complete symbol table for version 0.5

Symbol	**Definition**
ι	The Medium's act of self-distinction — produces character as the difference the
$\phi : M \to \{+1, -1\}$	Distinction character field — records characters produced by ι events
$x \sim y$	Co-manifestation — shared state, no spatial relationship
$\kappa = \sum \iota$	Coupling character — net distinction character of a local cluster
$= \sum \kappa$	Configuration energy — governs stability via extremum condition
$\hat{H} = \mathrm{proj}()$	Hamiltonian — projection of into spacetime
N	Count of ι events in a configuration — first approximation for mass
sgn()	Net distinction character of configuration energy — projects to charge
Spin	Relational closure of encounter pattern within the web
Adjacency constraint	Identical adjacent distinction characters forbidden — source of minimum thresh
Projection boundary	Planck scale — where dimensionless substrate meets dimensional observer
$h = \mathrm{proj}(\iota)$	Planck's constant — dimensional projection of one ι event
c	Projection-layer ratio of projected distance to projected time
G	Gravitational constant — substrate derivation not yet complete
$x \sim y \to$ entanglement	Co-manifestation projects to quantum entanglement
Entropy	Projection-layer emergent — observer's coarse-graining of microstates
Time	Projection-layer emergent — ordered account of substrate transitions
Redshift	Three-layer projection artifact — not recession velocity
Color charge	Not yet placed in either layer
Dark matter / dark energy	Substrate candidate structures not yet identified

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ORCID: https://orcid.org/0009-0001-0328-2299